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EBB 320

Control Systems

Practical 3: PID Controller Design

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1 Theoretical Background

1.1 First Order Responses

To find the rise time of a First Order Plus Dead Time (FOPDT) transfer function the following equation can be used: [1]

$$T_r = \frac{2.2}{a} \tag{1}$$

The dead time of the FOPDT transfer function can be approximated using Padé first order approximation:

$$e^{-\theta s} \approx \frac{1 - \frac{\theta}{2}s}{1 + \frac{\theta}{2}s} \tag{2}$$

1.2 Second Order Responses

The equation defining a standard second order response is defined as:

$$G(s) = \frac{{\omega_n}^2}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$$

The equations that then define the dampening ratio and natural radian frequency of the second order response using a pre-define overshoot and rise time are found to be: [1]

$$\zeta = \frac{-\ln(\%\mathrm{OS}/100)}{\sqrt{\pi^2 + \ln^2(\%\mathrm{OS}/100)}} \tag{3}$$

$$\begin{aligned} \omega_n T_r &= 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1 \\ \omega_n &= (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1)/T_r \end{aligned} \tag{4}$$

The equation defining the natural frequency using the system's rise time is found using curve fitting, as a precise analytical equation is not available. [1]

A design point associated with the found dampening ration and natural radian frequency can then be found using: [1]

$$\sigma_d = \zeta\omega_n \tag{5}$$

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} \tag{6}$$

1.3 PID Controllers

From [1] the basic equations that define a PID controller's transfer function are:

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_c(s + z_1)(s + z_2)}{s} \tag{7}$$

Therefore, it can be found that:

$$K_p = K_c(z_1 + z_2) \tag{8}$$
$$K_i = K_c(z_1 z_2) \tag{9}$$
$$K_d = K_c \tag{10}$$

To find the parameters of the PID controller the following equations must be satisfied in order to ensure system stability.

The angle equation:

$$\sum \text{zero angles} - \sum \text{pole angles} = \pm 180^\circ \tag{11}$$

The magnitude equation:

$$|K_c G_c(s_d) G_p(s_d)| = 1 \tag{12}$$

2 Design and Simulation

The following table contains the derived transfer functions that were used to simulate and implement the controllers.

Table 1: 2x2 Process Transfer Function Matrix

	Input 1 (U_1)	**Input 2** (U_2)
Output 1 (Y_1)	$G_{11}(s) = \frac{0.1179}{s+0.0076}e^{-0.145s}$	$G_{12}(s) = \frac{0.03658}{s+0.005254}e^{-0.145s}$
Output 2 (Y_2)	$G_{21}(s) = \frac{0.06368}{s+0.006012}e^{-0.145s}$	$G_{22}(s) = \frac{0.05872}{s+0.005984}e^{-0.145s}$

2.1 PID Controller Design for $G_{11}(s)$

The design specifications are a 10% overshoot and a rise time of $T_r = 96.1$ s. The damping ratio ζ is calculated as:

$$\zeta = \frac{-\ln(0.1)}{\sqrt{\pi^2 + [\ln(0.1)]^2}} = 0.591$$

The required natural frequency ω_n is:

$$\omega_n = \frac{1.76(0.591)^3 - 0.417(0.591)^2 + 1.039(0.591) + 1}{96.1} = 0.01906 \text{ rad/s}$$

The required closed-loop bandwidth ω_{BW} and phase margin Φ_M are:

$$\omega_{BW} = \omega_n\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} = 0.02211 \text{ rad/s}$$

$$\Phi_M = \tan^{-1}\left(\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{1+4\zeta^4}}}\right) = 58.58^\circ$$

Following the prescribed procedure, the PI controller zero is fixed at $z_2 = 0.1$, yielding $G_{PI}(s) = \frac{s+0.1}{s}$. The new phase-margin frequency ω_s is selected as 80% of the bandwidth:

$$\omega_s = 0.8 \times 0.02211 = 0.01769 \text{ rad/s}$$

The target total loop phase at ω_s is:

$$\phi_{required}(\omega_s) = -180^\circ + 58.58^\circ = -121.42^\circ$$

The phase contributions of the uncompensated plant (including dead time) and the PI controller at ω_s are:

$$\phi_G(0.01769) = -0.07^\circ - 66.77^\circ = -66.84^\circ$$

$$\phi_{PI}(0.01769) = -90^\circ + \tan^{-1}\left(\frac{0.01769}{0.1}\right) = -79.97^\circ$$

The required phase addition from the D-action zero is:

$$\phi_{add}(\omega_s) = -121.42^\circ - (-66.84^\circ) - (-79.97^\circ) = +25.39^\circ$$

The D-action zero z_1 is located to provide exactly 25.39° of phase:

$$\tan(25.39^\circ) = \frac{0.01769}{z_1} \implies z_1 = 0.03727$$

The controller gain K_{cPID} is calculated to force unity magnitude at ω_s:

$$|G_{11}(j0.01769)| \times \frac{K_{cPID}|j0.01769 + 0.03727||j0.01769 + 0.1|}{|j0.01769|} = 1$$

$$6.121 \times K_{cPID} \times \frac{0.04126 \times 0.10155}{0.01769} = 1 \implies K_{cPID} = 0.6901$$

Expanding the controller transfer function yields the parallel PID gains:

$$G_{c11}(s) = 0.6901\frac{(s + 0.03727)(s + 0.1)}{s} = 0.6901s + 0.0947 + \frac{0.00257}{s}$$

Final Gains for G_{11}: $K_p = 0.0947$, $K_i = 0.00257$, $K_d = 0.6901$

2.2 PID Controller Design for $G_{22}(s)$

For a target rise time of $T_r = 122.55$ s and identical 10% overshoot ($\zeta = 0.591$, $\Phi_M = 58.58^\circ$), the bandwidth requirements shift:

$$\omega_n = \frac{1.8316}{122.55} = 0.01495 \text{ rad/s}$$
$$\omega_{BW} = 0.01495 \times 1.160 = 0.01734 \text{ rad/s}$$

With $z_2 = 0.1$, the new phase-margin frequency is:

$$\omega_s = 0.8 \times 0.01734 = 0.01387 \text{ rad/s}$$

The required total loop phase is identical: $\phi_{required} = -121.42^\circ$. Evaluating the G_{22} plant and PI controller at the new ω_s:

$$\phi_G(0.01387) = -66.61^\circ(\text{Core}) - 0.12^\circ(\text{Delay}) = -66.73^\circ$$
$$\phi_{PI}(0.01387) = -90^\circ + \tan^{-1}\left(\frac{0.01387}{0.1}\right) = -82.10^\circ$$

The required phase addition is:

$$\phi_{add}(\omega_s) = -121.42^\circ - (-66.73^\circ) - (-82.10^\circ) = +27.41^\circ$$

The D-action zero z_1 is located to provide 27.41° of phase:

$$\tan(27.41^\circ) = \frac{0.01387}{z_1} \implies z_1 = 0.02674$$

The overall controller gain K_{cPID} is found via unity magnitude mapping:

$$|G_{22}(j0.01387)| \times \frac{K_{cPID}|j0.01387 + 0.02674||j0.01387 + 0.1|}{|j0.01387|} = 1$$

$$3.885 \times K_{cPID} \times \frac{0.03013 \times 0.10096}{0.01387} = 1 \implies K_{cPID} = 1.174$$

Expanding the controller polynomial yields:

$$G_{c22}(s) = 1.174\frac{(s + 0.02674)(s + 0.1)}{s} = 1.174s + 0.1488 + \frac{0.00314}{s}$$

Final Gains for G_{22}: $K_p = 0.1488$, $K_i = 0.00314$, $K_d = 1.174$

2.3 Root Locus Technique for G11

For the transfer function G_{11} using equation [1] the current rise time was found to be ≈ 289.78 sec. Therefore, the target rise time is 96.593 sec.

Then using equations [3] and [4] it was found that:

$$\begin{aligned}\zeta &\approx 0.5912\\ \omega_n &\approx 0.018967\end{aligned}$$

The design point was then found using [5] and [6] giving $s_d = -0.01121 + j0.0153$.

The second form of the PID equation [7] was then used to find the controller transfer function.

Furthermore, the Padé approximation [2] was used to find an approximation of the transfer function's dead time. The new plant transfer function was then found to be:

$$G_p(s) = \frac{(1 - 0.0725s)}{(1 + 0.0725s)} \frac{0.1175}{(s + 0.007592)}$$

Letting $z_2 = 0.1$ the angles of the poles and zeros to the design point, and [11], it was found:

$$\begin{aligned}&\theta_{z=z_1} + \theta_{z=z_2} + \theta_{z=13.793} - (\theta_{p=0} + \theta_{p=-13.793} + \theta_{p=-0.007592}) = \pm 180^\circ\\ &\theta_{z=z_1} \approx -180^\circ + 219.90^\circ\\ &\theta_{z=z_1} \approx 39.905^\circ\end{aligned}$$

Therefore, the value of z_1 was found as:

$$\begin{aligned}\tan(\theta_{z=z_1}) &= \frac{0.0153}{z_1 - 0.01121}\\ z_1 &= \frac{0.0153 + 0.01121\tan(\theta_{z=z_1})}{\tan(\theta_{z=z_1})}\\ z_1 &\approx 0.0295055\end{aligned}$$

The controller gain K_c was found using [12] where:

$$\begin{aligned}|K_c G_c(s_d) G_p(s_d)| &= 1\\ |G_c(s_d) G_p(s_d)| &\approx 0.848\end{aligned}$$

Therefore:

$$K_c = \frac{1}{|G_c(s_d) G_p(s_d)|} \approx 1.17921$$

The final controller transfer function was then found to be:

$$G_c(s) = \frac{1.17921(s + 0.1)(s + 0.0295055)}{s}$$

Using [10], [9] and [8] the PID values where found to be:

$$K_p = K_c(z_1 + z_2) = 0.152714$$
$$K_i = K_c(z_1 z_2) = 0.003479$$
$$K_d = K_c = 1.17921$$

In order to confirm the design, the root locus was made using matlab, where the root locus plot has an intersection very close to the design point.

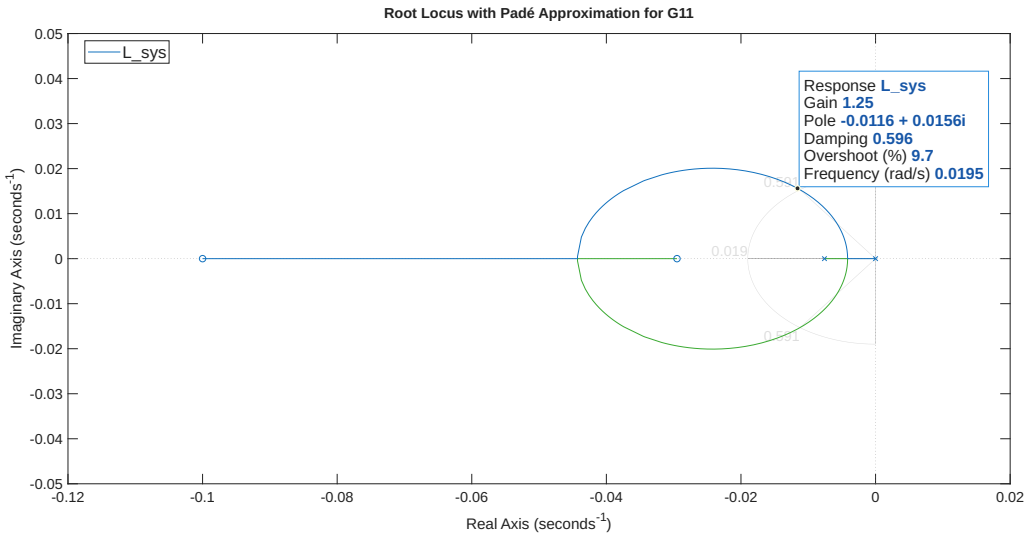


Figure 1: Root Locus plot of G11 with PID controller showing the intersection with the design point.

The resulting time domain comparison of the new closed loop transfer function and the open loop transfer function are shown below:

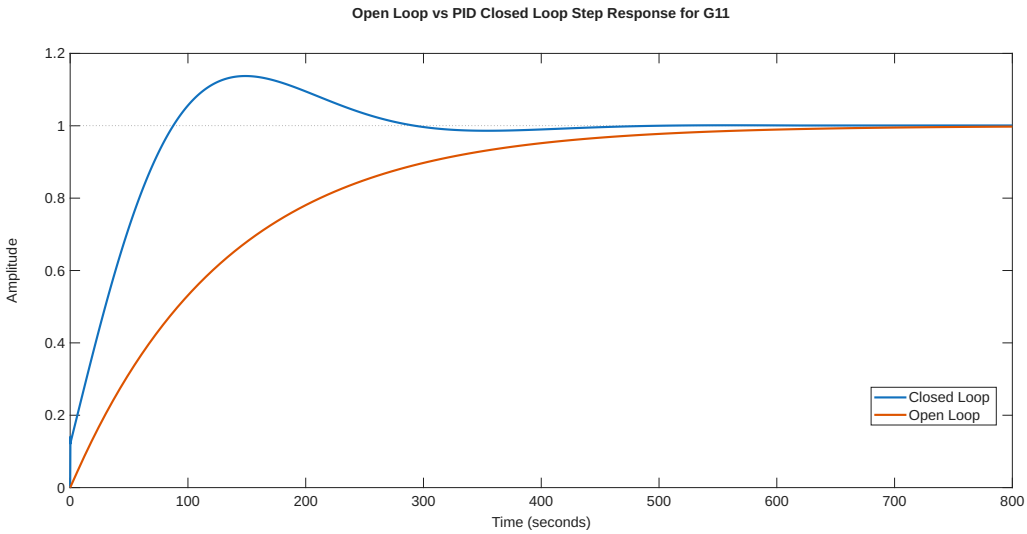


Figure 2: Time domain comparison of open loop plant transfer function against the closed loop system transfer function for G11

The following table compares the open loop and closed loop step response information.

Table 2: Table comparing open vs closed loop step response characteristics of G11

Step Response Parameter	Open Loop	Closed Loop
Rise Time (s)	289.39	71.09
Transient Time (s)	515.43	263.99
Settling Time (s)	515.43	263.99
Overshoot (%)	0.00	13.75
Undershoot (%)	-0.00	-0.00
Peak Time (s)	1025.12	148.79

2.4 Root Locus Technique for G22

For the transfer function G_{22} using equation [1] the current rise time was found to be ≈ 367.65 sec. Therefore, the target rise time is 122.55 sec.

Then using equations [3] and [4] it was found that:

$$\zeta \approx 0.5912$$
$$\omega_n \approx 0.01495$$

The design point was then found using [5] and [6] giving $s_d = -0.008838 + j0.0121$.

The second form of the PID equation [7] was then used to find the controller transfer function.

Furthermore, the Padé approximation [2] was used to find an approximation of the transfer function's dead time. The new plant transfer function was then found to be:

$$G_p(s) = \frac{(1 - 0.0725s)}{(1 + 0.0725s)} \frac{0.05872}{(s + 0.005984)}$$

Letting $z_2 = 0.1$ the angles of the poles and zeros to the design point, and [11], it was found:

$$\theta_{z=z_1} + \theta_{z=z_2} + \theta_{z=13.793} - (\theta_{p=0} + \theta_{p=-13.793} + \theta_{p=-0.005984}) = \pm 180^\circ$$
$$\theta_{z=z_1} \approx -180^\circ + 222.12^\circ$$
$$\theta_{z=z_1} \approx 42.119^\circ$$

Therefore, the value of z_1 was found as:

$$\tan(\theta_{z=z_1}) = \frac{0.0121}{z_1 - 0.00884}$$
$$z_1 = \frac{0.0121 + 0.00884 \tan(\theta_{z=z_1})}{\tan(\theta_{z=z_1})}$$
$$z_1 \approx 0.0221733$$

The controller gain K_c was found using [12] where:

$$|K_c G_c(s_d) G_p(s_d)| = 1$$
$$|G_c(s_d) G_p(s_d)| \approx 0.52473$$

Therefore:

$$K_c = \frac{1}{|G_c(s_d) G_p(s_d)|} \approx 1.90571$$

The final controller transfer function was then found to be:

$$G_c(s) = \frac{1.90571(s + 0.1)(s + 0.0221733)}{s}$$

Using [10], [9] and [8] the PID values where found to be:

$$K_p = K_c(z_1 + z_2) = 0.232827$$
$$K_i = K_c(z_1 z_2) = 0.0042256$$
$$K_d = K_c = 1.90571$$

In order to confirm the design, the root locus was made using matlab, where the root locus plot has an intersection very close to the design point.

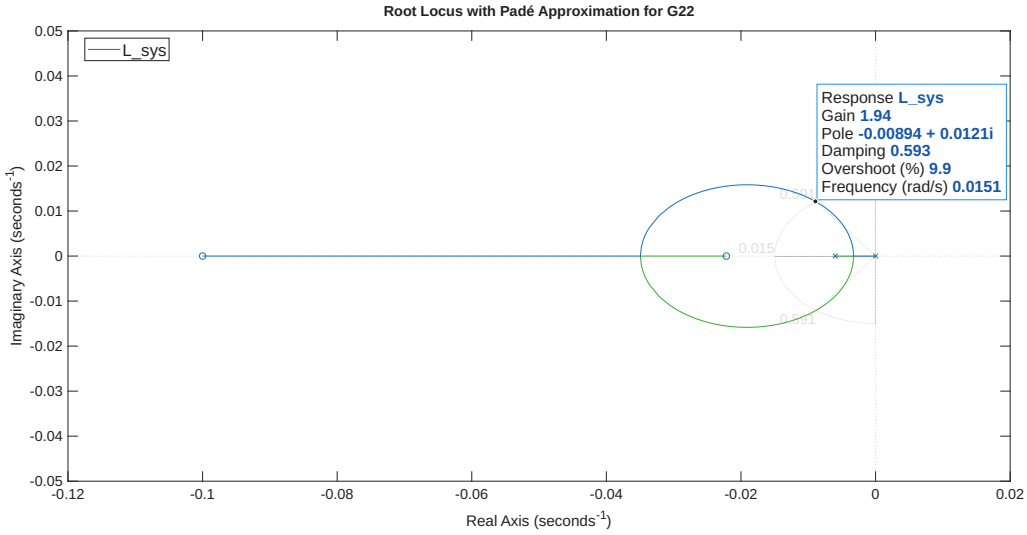


Figure 3: Root Locus plot of G22 with PID controller showing the intersection with the design point.

The resulting time domain comparison of the new closed loop transfer function and the open loop transfer function are shown below:

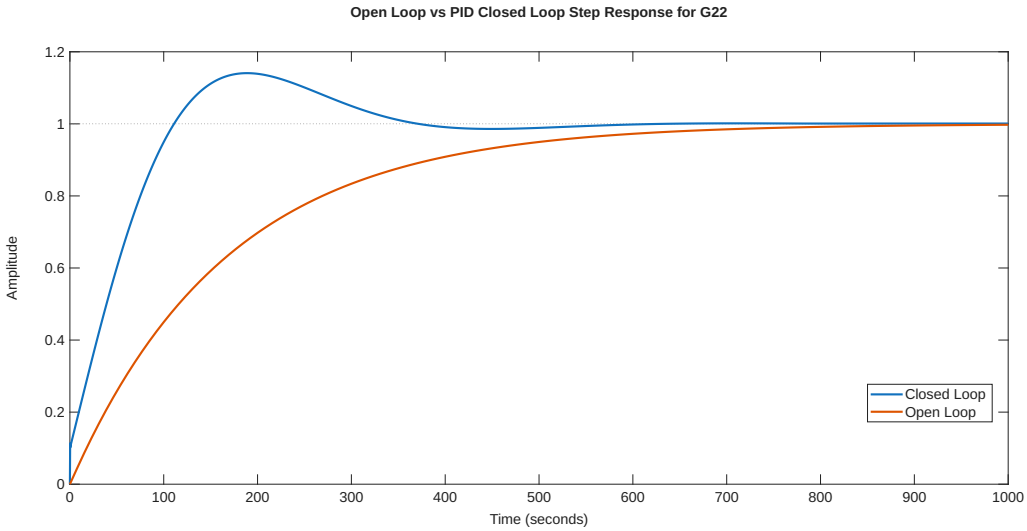


Figure 4: Time domain comparison of open loop plant transfer function against the closed loop system transfer function for G22

The following table compares the open loop and closed loop step response information.

Table 3: Table comparing open vs closed loop step response characteristics of G22

Step Response Parameter	Open Loop	Closed Loop
Rise Time (s)	367.15	90.64
Transient Time (s)	653.90	335.60
Settling Time (s)	653.90	335.60
Overshoot (%)	0.00	14.07
Undershoot (%)	-0.00	-0.00
Peak Time (s)	1300.59	188.79

3 Implementation

4 Results

5 Discussion

References

[1] N. S. Nise, *Control Systems Engineering*, 8th. John Wiley & Sons, 2019, ISBN: 978-1-119-47421-0.

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A Code

A.1 Matlab Code

Listing 1: App Code For Sampling

```
%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%% Auto Generated Code Start
    %%%%%%%%%%%%%%%%%
%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%


classdef EBBPrac < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        UIFigure                    matlab.ui.Figure
        GridLayout                  matlab.ui.container.
           GridLayout
        LeftPanel                   matlab.ui.container.
           Panel
        TabGroup                    matlab.ui.container.
           TabGroup
        Tab                         matlab.ui.container.
           Tab
        DataSaveEditField           matlab.ui.control.
           EditField
        DataSaveEditFieldLabel      matlab.ui.control.
           Label
        SamplingtoofastLabel        matlab.ui.control.
           Label
        UpdateChannelsButton        matlab.ui.control.
           Button
        StartButton                 matlab.ui.control.
           StateButton
        StopButton                  matlab.ui.control.
           StateButton
        OutputDropDown              matlab.ui.control.
           DropDown
        OutputDropDownLabel         matlab.ui.control.
           Label
        DutyCycleEditField          matlab.ui.control.
           NumericEditField
        DutyCycleEditFieldLabel     matlab.ui.control.
           Label
        DisturbanceEditField        matlab.ui.control.
```

```
        NumericEditField
    DisturbanceEditFieldLabel  matlab.ui.control.
        Label
    SetpointEditField          matlab.ui.control.
        NumericEditField
    SetpointEditFieldLabel     matlab.ui.control.
        Label
    DEditField                 matlab.ui.control.
        NumericEditField
    DEditFieldLabel            matlab.ui.control.
        Label
    IEditField                 matlab.ui.control.
        NumericEditField
    IEditFieldLabel            matlab.ui.control.
        Label
    PEditField                 matlab.ui.control.
        NumericEditField
    PEditFieldLabel            matlab.ui.control.
        Label
    Tab2                       matlab.ui.container.
        Tab
    BaudRateEditField          matlab.ui.control.
        NumericEditField
    BaudRateEditFieldLabel     matlab.ui.control.
        Label
    COMPortEditField           matlab.ui.control.
        EditField
    COMPortEditFieldLabel      matlab.ui.control.
        Label
    RightPanel                 matlab.ui.container.
        Panel
    UIAxes_2                   matlab.ui.control.
        UIAxes
    UIAxes                     matlab.ui.control.
        UIAxes
end

% Properties that correspond to apps with auto-reflow
properties (Access = private)
    onePanelWidth = 576;
end
%
   %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%% Auto Generated Code End
   %%%%%%%%%%%%%%%
%
   %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
properties (Access = private)
    %Live lines 1 and 2 are for the temperature
       measurements.
    liveLine1
    liveLine2
    %Live lines 3 and 4 are for the PWM signals.
    liveLine3
    liveLine4
    P
    I
    D
    setpoint
    disturbance
    duty_cycle
    output
    capturing
    COM_port
    baud_rate
    window_size
    esp32
    data_save_name % Description
end


% Callbacks that handle component events
methods (Access = private)

    % Value changed function: PEditField
    function PEditFieldValueChanged(app, event)
        app.P = app.PEditField.Value;
    end

    % Value changed function: IEditField
    function IEditFieldValueChanged(app, event)
        app.I = app.IEditField.Value;
    end

    % Value changed function: DEditField
    function DEditFieldValueChanged(app, event)
        app.D = app.DEditField.Value;
    end

    % Value changed function: SetpointEditField
    function SetpointEditFieldValueChanged(app, event
       )
        app.setpoint = app.SetpointEditField.Value;
```

```
end

% Value changed function: DisturbanceEditField
function DisturbanceEditFieldValueChanged(app,
    event)
    app.disturbance = app.DisturbanceEditField.
        Value;
end

% Value changed function: DutyCycleEditField
function DutyCycleEditFieldValueChanged(app,
    event)
    app.duty_cycle = app.DutyCycleEditField.Value
        ;
end

% Value changed function: OutputDropDown
function OutputDropDownValueChanged(app, event)
    app.output = str2double(app.OutputDropDown.
        Value);
end

% Value changed function: StartButton
function StartButtonValueChanged(app, event)
    app.StartButton.Visible = "off";
    app.StopButton.Visible = "on";
    app.UpdateChannelsButton.Visible = "on";
    app.capturing = true;

    %Call all of the other functions to make sure
        their values are
    %captured.
    app.PEditFieldValueChanged();
    app.IEditFieldValueChanged();
    app.DEditFieldValueChanged();
    app.SetpointEditFieldValueChanged();
    app.DisturbanceEditFieldValueChanged();
    app.DutyCycleEditFieldValueChanged();
    app.OutputDropDownValueChanged();
    app.BaudRateEditFieldValueChanged();
    app.COMPortEditFieldValueChanged();
    app.DataSaveEditFieldValueChanged();

    if isempty(app.esp32)
        app.esp32 = serialport(app.COM_port, app.
            baud_rate);
        configureTerminator(app.esp32, "LF");
    end
```

```
flush(app.esp32);

pause(1);

%Make sure all info is read into esp.
writeline(app.esp32, sprintf("%d, %.2f", 0,
   app.P)); %
writeline(app.esp32, sprintf("%d, %.2f", 1,
   app.I)); %
writeline(app.esp32, sprintf("%d, %.2f", 2,
   app.D)); %
writeline(app.esp32, sprintf("%d, %.2f", 3,
   app.setpoint)); %
writeline(app.esp32, sprintf("%d, %.2f", 4,
   app.disturbance)); %
writeline(app.esp32, sprintf("%d, %d", 5, app
   .duty_cycle)); %
writeline(app.esp32, sprintf("%d, %.2f", 6,
   app.output)); %
writeline(app.esp32, sprintf("%d, %.2f", 7,
   app.capturing)); %

%Clear axes.
cla(app.UIAxes);
cla(app.UIAxes_2);

%Plotting axes for the temperature
   measurements.
app.liveLine1 = animatedline(app.UIAxes, '
   Color', 'r');
app.liveLine2 = animatedline(app.UIAxes, '
   Color', 'b');

%Plotting axes for the PWM signals.
app.liveLine3 = animatedline(app.UIAxes_2, '
   Color', 'r');
app.liveLine4 = animatedline(app.UIAxes_2, '
   Color', 'b');

app.window_size = 30; %seconds

%Define data csv that we will safe to.
fid = fopen(app.data_save_name + ".csv", 'wt'
   );

while(app.capturing)
    if app.esp32.NumBytesAvailable > 0
        rawData = readline(app.esp32);
```

```
                data = str2double(split(rawData, ',')
                    );
                data(5) = data(5)/1000;

                if ~isnan(data(1)) && (length(data)
                    == 6)
                    %Plot temperature data.
                    addpoints(app.liveLine1, data(5),
                        data(1));
                    addpoints(app.liveLine2, data(5),
                        data(2));
                    xlim(app.UIAxes, [data(5) - app.
                        window_size, data(5)]);

                    %Plot PWM signals.
                    addpoints(app.liveLine3, data(5),
                        data(3));
                    addpoints(app.liveLine4, data(5),
                        data(4));
                    xlim(app.UIAxes_2, [data(5) - app
                        .window_size, data(5)]);

                    %Write data to file.
                    fprintf(fid, rawData);

                    if data(end) == 0
                        app.SamplingtoofastLabel.
                            Visible = "on";
                    else
                        app.SamplingtoofastLabel.
                            Visible = "off";
                    end
                end

                drawnow limitrate
            end
        end

        %Close file we are saving to.
        fclose(fid);

        %Send stop signal.
        writeline(app.esp32, sprintf("%d, %.2f", 7,
            app.capturing));

        clear app.esp32;
    end
```

```
% Value changed function: COMPortEditField
function COMPortEditFieldValueChanged(app, event)
    app.COM_port = app.COMPortEditField.Value;
end

% Value changed function: BaudRateEditField
function BaudRateEditFieldValueChanged(app, event
    )
    app.baud_rate = app.BaudRateEditField.Value;
end

% Value changed function: StopButton
function StopButtonValueChanged(app, event)
    app.capturing = false;
    app.StopButton.Visible = "off";
    app.UpdateChannelsButton.Visible = "off";
    app.StartButton.Visible = "on";

end

% Button pushed function: UpdateChannelsButton
function UpdateChannelsButtonPushed(app, event)
    app.DutyCycleEditFieldValueChanged();
    app.OutputDropDownValueChanged();
    writeline(app.esp32, sprintf("%d, %.2f", 5,
        app.duty_cycle));
    writeline(app.esp32, sprintf("%d, %.2f", 6,
        app.output));
end

% Value changed function: DataSaveEditField
function DataSaveEditFieldValueChanged(app, event
    )
    app.data_save_name = app.DataSaveEditField.
        Value;

end

% Changes arrangement of the app based on
    UIFigure width
function updateAppLayout(app, event)
    currentFigureWidth = app.UIFigure.Position(3)
        ;
    if(currentFigureWidth <= app.onePanelWidth)
        % Change to a 2x1 grid
        app.GridLayout.RowHeight = {449, 449};
        app.GridLayout.ColumnWidth = {'1x'};
        app.RightPanel.Layout.Row = 2;
```

```
            app.RightPanel.Layout.Column = 1;
        else
            % Change to a 1x2 grid
            app.GridLayout.RowHeight = {'1x'};
            app.GridLayout.ColumnWidth = {204, '1x'};
            app.RightPanel.Layout.Row = 1;
            app.RightPanel.Layout.Column = 2;
        end
    end
end

%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%% Auto Generated Code Start
    %%%%%%%%%%%%%
%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%


% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create UIFigure and hide until all
           components are created
        app.UIFigure = uifigure('Visible', 'off');
        app.UIFigure.AutoResizeChildren = 'off';
        app.UIFigure.Position = [100 100 702 449];
        app.UIFigure.Name = 'MATLAB App';
        app.UIFigure.SizeChangedFcn =
           createCallbackFcn(app, @updateAppLayout,
           true);

        % Create GridLayout
        app.GridLayout = uigridlayout(app.UIFigure);
        app.GridLayout.ColumnWidth = {204, '1x'};
        app.GridLayout.RowHeight = {'1x'};
        app.GridLayout.ColumnSpacing = 0;
        app.GridLayout.RowSpacing = 0;
        app.GridLayout.Padding = [0 0 0 0];
        app.GridLayout.Scrollable = 'on';

        % Create LeftPanel
        app.LeftPanel = uipanel(app.GridLayout);
        app.LeftPanel.Layout.Row = 1;
```

```
app.LeftPanel.Layout.Column = 1;

% Create TabGroup
app.TabGroup = uitabgroup(app.LeftPanel);
app.TabGroup.Position = [3 3 197 442];

% Create Tab
app.Tab = uitab(app.TabGroup);
app.Tab.Title = 'Tab';

% Create PEditFieldLabel
app.PEditFieldLabel = uilabel(app.Tab);
app.PEditFieldLabel.HorizontalAlignment = '
   right';
app.PEditFieldLabel.Position = [46 385 25
   22];
app.PEditFieldLabel.Text = 'P';

% Create PEditField
app.PEditField = uieditfield(app.Tab, '
   numeric');
app.PEditField.ValueChangedFcn =
   createCallbackFcn(app, @
   PEditFieldValueChanged, true);
app.PEditField.Position = [85 385 100 22];

% Create IEditFieldLabel
app.IEditFieldLabel = uilabel(app.Tab);
app.IEditFieldLabel.HorizontalAlignment = '
   right';
app.IEditFieldLabel.Position = [46 355 25
   22];
app.IEditFieldLabel.Text = 'I';

% Create IEditField
app.IEditField = uieditfield(app.Tab, '
   numeric');
app.IEditField.ValueChangedFcn =
   createCallbackFcn(app, @
   IEditFieldValueChanged, true);
app.IEditField.Position = [85 355 100 22];

% Create DEditFieldLabel
app.DEditFieldLabel = uilabel(app.Tab);
app.DEditFieldLabel.HorizontalAlignment = '
   right';
app.DEditFieldLabel.Position = [46 326 25
   22];
```

```
app.DEditFieldLabel.Text = 'D';

% Create DEditField
app.DEditField = uieditfield(app.Tab, '
   numeric');
app.DEditField.ValueChangedFcn =
   createCallbackFcn(app, @
   DEditFieldValueChanged, true);
app.DEditField.Position = [85 326 100 22];

% Create SetpointEditFieldLabel
app.SetpointEditFieldLabel = uilabel(app.Tab)
   ;
app.SetpointEditFieldLabel.
   HorizontalAlignment = 'right';
app.SetpointEditFieldLabel.Position = [21 296
    49 22];
app.SetpointEditFieldLabel.Text = 'Setpoint';

% Create SetpointEditField
app.SetpointEditField = uieditfield(app.Tab,
   'numeric');
app.SetpointEditField.ValueChangedFcn =
   createCallbackFcn(app, @
   SetpointEditFieldValueChanged, true);
app.SetpointEditField.Position = [85 296 100
   22];

% Create DisturbanceEditFieldLabel
app.DisturbanceEditFieldLabel = uilabel(app.
   Tab);
app.DisturbanceEditFieldLabel.
   HorizontalAlignment = 'right';
app.DisturbanceEditFieldLabel.Position = [1
   265 69 22];
app.DisturbanceEditFieldLabel.Text = '
   Disturbance';

% Create DisturbanceEditField
app.DisturbanceEditField = uieditfield(app.
   Tab, 'numeric');
app.DisturbanceEditField.ValueChangedFcn =
   createCallbackFcn(app, @
   DisturbanceEditFieldValueChanged, true);
app.DisturbanceEditField.Position = [85 265
   100 22];

% Create DutyCycleEditFieldLabel
```

```
app.DutyCycleEditFieldLabel = uilabel(app.Tab
    );
app.DutyCycleEditFieldLabel.
    HorizontalAlignment = 'right';
app.DutyCycleEditFieldLabel.Position = [9 235
     63 22];
app.DutyCycleEditFieldLabel.Text = 'Duty
    Cycle';

% Create DutyCycleEditField
app.DutyCycleEditField = uieditfield(app.Tab,
     'numeric');
app.DutyCycleEditField.ValueChangedFcn =
    createCallbackFcn(app, @
    DutyCycleEditFieldValueChanged, true);
app.DutyCycleEditField.Position = [86 235 100
     22];
app.DutyCycleEditField.Value = 100;

% Create OutputDropDownLabel
app.OutputDropDownLabel = uilabel(app.Tab);
app.OutputDropDownLabel.HorizontalAlignment =
     'right';
app.OutputDropDownLabel.Position = [30 199 41
     22];
app.OutputDropDownLabel.Text = 'Output';

% Create OutputDropDown
app.OutputDropDown = uidropdown(app.Tab);
app.OutputDropDown.Items = {'None', 'Channel1
    ', 'Channel2', 'Both'};
app.OutputDropDown.ItemsData = {'3', '0', '1'
    , '2'};
app.OutputDropDown.ValueChangedFcn =
    createCallbackFcn(app, @
    OutputDropDownValueChanged, true);
app.OutputDropDown.Position = [85 199 100
    22];
app.OutputDropDown.Value = '3';

% Create StopButton
app.StopButton = uibutton(app.Tab, 'state');
app.StopButton.ValueChangedFcn =
    createCallbackFcn(app, @
    StopButtonValueChanged, true);
app.StopButton.Visible = 'off';
app.StopButton.Text = 'Stop';
app.StopButton.Position = [85 168 100 23];
```

```
% Create StartButton
app.StartButton = uibutton(app.Tab, 'state');
app.StartButton.ValueChangedFcn =
   createCallbackFcn(app, @
   StartButtonValueChanged, true);
app.StartButton.Text = 'Start';
app.StartButton.Position = [86 168 100 23];

% Create UpdateChannelsButton
app.UpdateChannelsButton = uibutton(app.Tab,
   'push');
app.UpdateChannelsButton.ButtonPushedFcn =
   createCallbackFcn(app, @
   UpdateChannelsButtonPushed, true);
app.UpdateChannelsButton.Visible = 'off';
app.UpdateChannelsButton.Position = [81 136
   108 23];
app.UpdateChannelsButton.Text = 'Update
   Channels';

% Create SamplingtoofastLabel
app.SamplingtoofastLabel = uilabel(app.Tab);
app.SamplingtoofastLabel.Visible = 'off';
app.SamplingtoofastLabel.Position = [90 12
   101 22];
app.SamplingtoofastLabel.Text = 'Sampling too
    fast!';

% Create DataSaveEditFieldLabel
app.DataSaveEditFieldLabel = uilabel(app.Tab)
   ;
app.DataSaveEditFieldLabel.
   HorizontalAlignment = 'right';
app.DataSaveEditFieldLabel.Position = [10 96
   61 22];
app.DataSaveEditFieldLabel.Text = 'Data Save'
   ;

% Create DataSaveEditField
app.DataSaveEditField = uieditfield(app.Tab,
   'text');
app.DataSaveEditField.ValueChangedFcn =
   createCallbackFcn(app, @
   DataSaveEditFieldValueChanged, true);
app.DataSaveEditField.Position = [86 96 100
   22];
app.DataSaveEditField.Value = 'PWM1StepV0';
```

```
% Create Tab2
app.Tab2 = uitab(app.TabGroup);
app.Tab2.Title = 'Tab2';

% Create COMPortEditFieldLabel
app.COMPortEditFieldLabel = uilabel(app.Tab2)
    ;
app.COMPortEditFieldLabel.HorizontalAlignment
     = 'right';
app.COMPortEditFieldLabel.Position = [13 385
    58 22];
app.COMPortEditFieldLabel.Text = 'COM Port';

% Create COMPortEditField
app.COMPortEditField = uieditfield(app.Tab2,
    'text');
app.COMPortEditField.ValueChangedFcn =
    createCallbackFcn(app, @
    COMPortEditFieldValueChanged, true);
app.COMPortEditField.Position = [85 385 100
    22];
app.COMPortEditField.Value = '/dev/ttyUSB0';

% Create BaudRateEditFieldLabel
app.BaudRateEditFieldLabel = uilabel(app.Tab2
    );
app.BaudRateEditFieldLabel.
    HorizontalAlignment = 'right';
app.BaudRateEditFieldLabel.Position = [9 355
    62 22];
app.BaudRateEditFieldLabel.Text = 'Baud Rate'
    ;

% Create BaudRateEditField
app.BaudRateEditField = uieditfield(app.Tab2,
     'numeric');
app.BaudRateEditField.ValueChangedFcn =
    createCallbackFcn(app, @
    BaudRateEditFieldValueChanged, true);
app.BaudRateEditField.Position = [85 355 100
    22];
app.BaudRateEditField.Value = 115200;

% Create RightPanel
app.RightPanel = uipanel(app.GridLayout);
app.RightPanel.Layout.Row = 1;
app.RightPanel.Layout.Column = 2;
```

```
            % Create UIAxes
            app.UIAxes = uiaxes(app.RightPanel);
            title(app.UIAxes, 'Title')
            xlabel(app.UIAxes, 'X')
            ylabel(app.UIAxes, 'Y')
            zlabel(app.UIAxes, 'Z')
            app.UIAxes.Position = [7 213 489 235];

            % Create UIAxes_2
            app.UIAxes_2 = uiaxes(app.RightPanel);
            title(app.UIAxes_2, 'Title')
            xlabel(app.UIAxes_2, 'X')
            ylabel(app.UIAxes_2, 'Y')
            zlabel(app.UIAxes_2, 'Z')
            app.UIAxes_2.Position = [8 7 488 207];

            % Show the figure after all components are
               created
            app.UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = EBBPrac

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.UIFigure)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.UIFigure)
        end
    end
end
```

```
%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%% Auto Generated Code end
    %%%%%%%%%%%%%
%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

A.2 Model Fit Code

Listing 2: Model fitting and plotting code

```
% Obtaining the transfer function
% Package the input arrays of both transistors into one
    matrix (Needs to be columns)
PWM1StepV1Data = readmatrix('..\Data1\Step_PWM1_V0.csv');
PWM1StepV1 = PWM1StepV1Data(7920:end,1:5);
PWM2StepV0Data = readmatrix('..\Data1\Step_PWM2_V0.csv');
PWM2StepV0 = PWM2StepV0Data(6400:end,1:5);

PWM1TestDataFile = readmatrix('..\Data1\
    Step_PWM1_VtestData50.csv');
PWM1TestData = PWM1TestDataFile(6230:end,1:5);
PWM2TestDataFile = readmatrix('..\Data1\
    Step_PWM2_VtestData50.csv');
PWM2TestData = PWM2TestDataFile(6160:end,1:5);

% pwm1 data prep
V_PWM1 = [PWM1StepV1(:,3)];
Temp_PWM1 = [PWM1StepV1(:,1)-22.28, PWM1StepV1(:,2)
    -21.24];
time_step = 0.005;

% pwm2 data prep
V_PWM2 = [PWM2StepV0(:,4)];
Temp_PWM2 = [PWM2StepV0(:,1)-20.26, PWM2StepV0(:,2)
    -19.05];

% pwm1 test data prep
V_PWM1_Test = [PWM1TestData(:,3)];
Temp_PWM1_test = [PWM1TestData(:,1)-21.2, PWM1TestData
    (:,2)-20.2];

% pwm2 test data prep
V_PWM2_Test = [PWM2TestData(:,4)];
Temp_PWM2_test = [PWM2TestData(:,1)-21.1, PWM2TestData
    (:,2)-20.2] ;
```

```
% use iddata to set up the data for tfest
tfest_data1 = iddata(Temp_PWM1, V_PWM1, time_step);
tfest_data2 = iddata(Temp_PWM2, V_PWM2, time_step);
testData1 = iddata(Temp_PWM1_test, V_PWM1_Test, time_step
    );
testData2 = iddata(Temp_PWM2_test, V_PWM2_Test, time_step
    );

% estimate nr of poles and zeros
n_poles = 1;
n_zeros = 0;

% Estimate the transfer functions independently (SISO) to
    avoid idss error
% tfest_data(samples, output, input)
sys11 = tfest(tfest_data1(:, 1, 1), n_poles, n_zeros, NaN
    ); % PWM1 to T1
sys21 = tfest(tfest_data1(:, 2, 1), n_poles, n_zeros, NaN
    ); % PWM1 to T2

sys12 = tfest(tfest_data2(:, 1, 1), n_poles, n_zeros, NaN
    ); % PWM2 to T1
sys22 = tfest(tfest_data2(:, 2, 1), n_poles, n_zeros, NaN
    ); % PWM2 to T2

% Combine the individual SISO models back into SIMO
    systems
simo_transfer1 = [sys11; sys21];
simo_transfer2 = [sys12; sys22];

% set up graph details
tfest_data1.Name = 'Measured Data';
tfest_data2.Name = 'Measured Data';
testData1.Name = 'Measured Data';
testData2.Name = 'Measured Data';
simo_transfer2.Name = 'Simulated Model';
simo_transfer1.Name = 'Simulated Model';

figure(1)
compare(tfest_data1, simo_transfer1);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

figure(2)
compare(tfest_data2, simo_transfer2);
legend('Location', 'best');
ylabel('Change in Temperature (C)')
```

```
figure(3)
compare(testData1, simo_transfer1);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

figure(4)
compare(testData2, simo_transfer2);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

lines = findobj(gcf, 'Type', 'Line');
set(lines, 'LineWidth', 1.5);

% display the transfer functions of PWM1
[num1, den1] = tfdata(simo_transfer1(1, 1), 'v');
delay1 = simo_transfer1(1, 1).IODelay; % Extract dead
   time
[num2, den2] = tfdata(simo_transfer1(2, 1), 'v');
delay2 = simo_transfer1(2, 1).IODelay;

disp('--- Transfer Function 1 (PWM1 to T1) ---');
disp('Numerator:'); disp(num1);
disp('Denominator:'); disp(den1);
disp(['Estimated Dead Time (s): ', num2str(delay1)]);

disp('--- Transfer Function 2 (PWM1 to T2) ---');
disp('Numerator:'); disp(num2);
disp('Denominator:'); disp(den2);
disp(['Estimated Dead Time (s): ', num2str(delay2)]);

% display transfer functions of PWM2
[num12, den12] = tfdata(simo_transfer2(1, 1), 'v');
delay12 = simo_transfer2(1, 1).IODelay;
[num22, den22] = tfdata(simo_transfer2(2, 1), 'v');
delay22 = simo_transfer2(2, 1).IODelay;

disp('--- Transfer Function 1 (PWM2 to T1) ---');
disp('Numerator:'); disp(num12);
disp('Denominator:'); disp(den12);
disp(['Estimated Dead Time (s): ', num2str(delay12)]);

disp('--- Transfer Function 2 (PWM2 to T2) ---');
disp('Numerator:'); disp(num22);
disp('Denominator:'); disp(den22);
disp(['Estimated Dead Time (s): ', num2str(delay22)]);
```

A.3 Prac 2 — Controller Design (P and PI)

A.4 Prac 2 — Report Figure Generation (P and PI)

A.5 MCU Code

```
#include <fmt/core.h>
#include <Arduino.h>
#include <iostream>
#include <sstream>
#include <vector>
#include <string>
#include <queue>
#include "ValuesPID.cpp"


double duty1 = 128; //50% duty cycle by default.
double duty2 = 128;
int control_outputs = 0; //By default only pwm1 is on.
bool start = 0; //Start off.
int message = 100; //Flags for certain messages.
int start_time = 0;
bool startControlling = 0; //See if we should start
    controlling, or continue just measuring.




//New PID structs for eventual use in the controllers.
ValuesPID controllerG11 = {0, 0, 0, 0, 0, 0, 0, 0};
ValuesPID controllerG22 = {0, 0, 0, 0, 0, 0, 0, 0};


double temp1 = 0; //Final temperature measurement that is
    sent to PC and used by controller/s.
double temp2 = 0; //Final temperature measurement that is
    sent to PC and used by controller/s.
int N_samples = 200; //Amount of samples being averaged.
std::queue<double> samples1; //Storage of samples for temp
    measurements for T1
std::queue<double> samples2; //Storage of samples for temp
    measurements for T2

hw_timer_t *timer = NULL; //Timer to ensure constant sample
    rate.
const int PWM_Out_1 = GPIO_NUM_32;
const int PWM_Out_2 = GPIO_NUM_25;
const int PWM_Meas_1 = GPIO_NUM_33;
const int PWM_Meas_2 = GPIO_NUM_26;
```

```
const int Temp_Meas1 = GPIO_NUM_27;
const int Temp_Meas2 = GPIO_NUM_35;

const double A = (3.3/4095.0); //Constant multiplier used to
    scale PWM outputs to voltages.
const int PWM_freq = 10;
const int PWM_res = 8;

const int BaudRate = 1152000;
const int sampling_period = 5000; //In microseconds.
const float dt = sampling_period/1000000.0; //t delta used
   in PWM calculations.
bool first_run = true; //Check if this is the very first run
   , used the prevent initial derivative kick. (Can be used
   for both controllers at once since they are controlling
   at the same time?)

float calculatePID(ValuesPID& controller){

      float duty = 0;
      float next_integral = controller.integral + controller
         .error * dt;

      controller.derivative = (controller.error - controller
         .prev_error)/dt;

      float pid_out = controller.P * controller.error +
         controller.I*next_integral + controller.D*
         controller.derivative;
      //anti-windup logic
      if  (pid_out >= 3.3 )
      {
        duty = 1;
        if ( controller.error < 0){
        controller.integral = next_integral;
        }

      }

      else if (pid_out <= 0.0 ) {
      duty = 0.0; // You cannot have a negative duty cycle
         for a simple heater/cooler
      if ( controller.error > 0){
        controller.integral = next_integral;
        }

      }
```

```
        else{
          duty = pid_out/3.3;
          controller.integral = next_integral;
        }
        duty = (pow(2, PWM_res) - 1)*(duty);

    return duty;
}

void setup() {
  //Start the USB-C serial/UART port.
  Serial.begin(BaudRate);

  //Setup I/O pins.
  pinMode(Temp_Meas1, INPUT);
  pinMode(Temp_Meas2, INPUT);
  pinMode(PWM_Meas_1, INPUT);
  pinMode(PWM_Meas_2, INPUT);

  ledcChangeFrequency(0, PWM_freq, PWM_res);
  ledcChangeFrequency(1, PWM_freq, PWM_res);
  ledcAttachPin (PWM_Out_1, 0);
  ledcAttachPin(PWM_Out_2, 1);

  timer = timerBegin(0, 80, true);

  samples1.push(0);
  samples2.push(0);
}

void loop() {

  //If receiving communication from computer, update values.
  if(Serial.available() > 0)
  {
    //Read the entire line, trim whitespace, and separate
       the comma separated line into the command value pair.
    String received = Serial.readStringUntil('\n');
    received.trim();

    std::istringstream stream(received.c_str());
    std::string command;
    std::string value_str;
    double value;

    std::getline(stream, command, ',');
    std::getline(stream, value_str, ',');
```

```
value = std::stod(value_str);

//Essentially a "switch case" to update the appropriate
    values for each command.
if(command == "0") // P value for PWM1
{
  controllerG11.P = value;
}
else if(command == "1") // I value for PWM1
{
  controllerG11.I = value;
}
else if(command == "2") // D value for PWM1
{
  controllerG11.D = value;
}
else if(command == "3") // Setpoint value for PWM1
{
  controllerG11.setpoint = value;
}

else if(command == "5") //Change from duty cycle set to
    the startControlling flag.
{
  startControlling = value;
}
else if(command == "7") //Flag to start process of
    taking measurements and sending them over UART.
{
  start = value;

  //Check if start is true, then start timer.
  if(start)
  {
    start_time = millis();
  }
}

else if(command == "8") // P value for PWM2
{
  controllerG22.P = value;
}
else if(command == "9") // I value for PWM2
{
  controllerG22.I = value;
}
else if(command == "10") // D value for PWM2
{
```

```
      controllerG22.D = value;
    }
    else if(command == "11") // Setpoint value for PWM2
    {
      controllerG22.setpoint = value;
    }
  }

  //If start flag is true, start measuring inputs/start
     controlling aswell.
  if(start)
  {
    //Check when we started in order to see how long we were
        busy.
    timerRestart(timer);

    if(startControlling)
    {
      //Turn on the appropriate PWM signals.
        ledcWrite(0, duty1);
        ledcWrite(1, duty2);
    }

    //Interpret sensor data.
    double temp1_raw = (analogReadMilliVolts(Temp_Meas1)
       /1000.0 - 0.5)/0.01;
    double temp2_raw = (analogReadMilliVolts(Temp_Meas2)
       /1000.0 - 0.5)/0.01;

    if(samples1.size() > N_samples)
    {
      temp1 -= samples1.front();
      temp2 -= samples2.front();

      samples1.pop();
      samples2.pop();
    }

    temp1 += temp1_raw/N_samples;
    temp2 += temp2_raw/N_samples;

    samples1.push(temp1_raw/N_samples);
    samples2.push(temp2_raw/N_samples);

    //Send sensor input data as a comma seperated list.
    std::string sending = fmt::format("{0},␣{1},␣{2},␣{3},␣
       {4},␣{5}",
    temp1, temp2, analogRead(PWM_Meas_1)*A, analogRead(
```

```
   PWM_Meas_2)*A, millis() - start_time, message);

Serial.println(String(sending.c_str()));
Serial.flush();


if (startControlling)
{
  //PID controller start
  controllerG11.prev_error = controllerG11.error;
  controllerG11.error = controllerG11.setpoint - temp1;

  controllerG22.prev_error =  controllerG22.error;
  controllerG22.error =  controllerG22.setpoint - temp2;

  // Prevent derivative kick on first run

  if (first_run) {
    controllerG11.prev_error = controllerG11.error;
    controllerG22.prev_error = controllerG22.error;

    first_run = false;
  }

  duty1 = calculatePID(controllerG11);
  duty2 =calculatePID(controllerG22);



  // std::string sending = fmt::format("{0}, {1}, {2},
     {3}, {4}", P, I, D, error, duty);
  // Serial.println(String(sending.c_str()));
  // Serial.flush();
}
//Otherwise turn off control signals.
else
{
  ledcWrite(0, 0);
  ledcWrite(1, 0);
}

//Check how long everything took in microseconds.
int elapsed = timerReadMicros(timer);

if(elapsed <= sampling_period)
{
  delayMicroseconds(sampling_period - elapsed);
  message = 100; //Reset message flag to show nothing.
```

```
      }
      else
      {
        delayMicroseconds(sampling_period);
        message = 0; //Set message flag for sampling too fast.
      }


    //Otherwise deactivate PWM outputs.
    }
    else
    {
      ledcWrite(0, 0);
      ledcWrite(1, 0);
    }
}
```